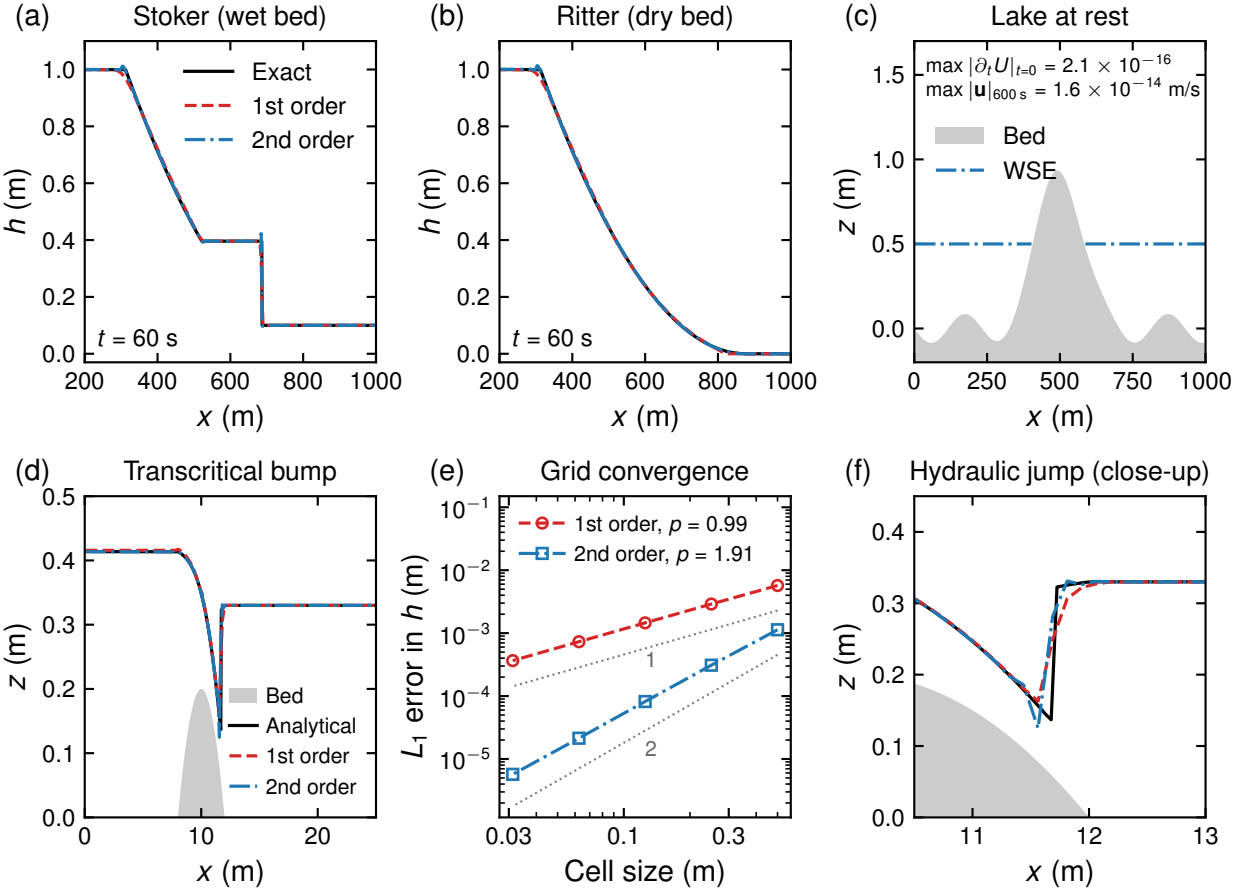

(a)
Stoker (wet bed)
Exact
1st order
2nd order
h (m)
t = 60 s
x (m)
(b)
Ritter (dry bed)
h (m)
t = 60 s
x (m)
(c)
Lake at rest
max |∂t U|t=0 = 2.1 × 10−16
max |u|600 s = 1.6 × 10−14 m/s
Bed
WSE
z (m)
x (m)
(d)
Transcritical bump
Bed
Analytical
1st order
2nd order
z (m)
x (m)
(e)
Grid convergence
1st order, p = 0.99
2nd order, p = 1.91
L1 error in h (m)
1
2
Cell size (m)
(f)
Hydraulic jump (close-up)
z (m)
x (m)